Predicting for Punctuality: Visualizing US Airline Delays

1. Introduction

Flight delays are an increasingly pertinent issue in modern air travel, driven by growing passenger demand, operational complexity, and external disruptions. U.S. Bureau of Transportation Statistics (BTS, n.d.) records show that arrival delay rates have risen from 17% in 2016 to 22% in 2025 and are expected to continue increasing. Common culprits are late-arriving aircraft, National Aviation System (NAS) constraints, air carrier issues, extreme weather, and security (BTS, 2024; Li & Yao, 2025; CMAC, 2022).

The costs are significant as airline delays cost the U.S. an estimated $32.9 billion in 2007 in operating expenses, lost passenger time, and reduced GDP (Ball et al., 2010). Research also shows that delays can propagate across airline networks, where disruptions at one airport can cascade through the system (Li & Yao, 2025). Climate-related weather disruptions are rising to nearly 30% in 2023, increasing unpredictability (IATA, 2024). Beyond economic costs, delays also affect passenger satisfaction (Oliveira et al., 2023), airline compensation liabilities (Zámková et al., 2022), and fuel consumption (Ryerson et al., 2014).

Most existing tools to predict flight delays focus on the individual flight level, without considering the structural context of the airports and routes involved. This project aims to move beyond single-flight prediction by incorporating network-derived features that capture an airport's connectivity and role within the aviation system. By integrating machine learning models with interactive visualisations, the system enables users to explore delay patterns across routes and airports, and make more informed travel decisions.

2. Problem definition

A flight delay is formally defined as an arrival occurring more than 15 minutes after the scheduled time (Li & Yao, 2025). Between 2000 and 2019, passenger traffic revenue increased at an annual rate of 5.5% (Sion et al., 2025), and the sustained growth continued to intensify operational complexity, strain capacity and contribute structurally to flight delay vulnerability. Current flight delay prediction systems primarily operate at the level of individual flights, focusing on whether a specific flight will be delayed based on historical and operational features. While these approaches have demonstrated strong predictive performance, they fail to account for the structural context of the network — such as how connected or central an airport is, and how these characteristics relate to delay risk. As flight networks become increasingly complex, incorporating this network-level context is important for both operational planning and travel decision-making.

We define the problem as follows: Given a flight dataset with features such as origin, destination, time and historical delays, the goal is to predict whether a flight or route will be delayed (binary classification) and the expected delay duration (regression). In addition, the aviation system is modelled as a network where airports are nodes and routes are edges, from which structural features such as node degree and betweenness centrality are extracted to enrich the predictive models. The objective of this project is to design and implement a system that incorporates network-level context into flight delay prediction, rather than treating flights in isolation, and to create interactive visualisations that enable travellers, airlines and airports to explore delay patterns and make more informed decisions.

3. Literature Survey

Research into data-driven flight delay prediction has grown substantially over the past decade. Li and Yao (2025) concluded that while machine learning methods handle the prediction of cascading, network-wide delays within individual airlines well, deep learning models that account for both geographic and temporal patterns are better suited to the more complex problem of propagation across airport networks. Individual studies illustrate how widely these methods have been applied: Hatıpoğlu and Tosun (2024) found XGBoost to be the strongest of seven models tested on Turkish airport data at 80% accuracy, while Tang (2022) reached 97.7% accuracy using a decision tree on JFK departure data. Kim and Park (2024) extended this across three international airports, finding accuracy remained largely stable from two to 48 hours ahead — suggesting long-range forecasting is more viable than assumed.

A parallel strand of research has focused on pushing predictive performance further and connecting it to real operational problems. Dai (2024) achieved 97.2% accuracy with a hybrid clustering and random forest approach, also finding that decomposing the problem improved processing speed by nearly 40%. AlBassam and AlShahrani (2025) demonstrated that data preparation — specifically correcting for the imbalance between on-time and delayed flights — matters as much as algorithm choice, with Random Forest and Decision Tree both reaching 90% accuracy once resampling was applied. Most concretely, Cao et al. (2024) fed arrival time predictions directly into a gate assignment framework in place of scheduled times, reducing potential gate conflicts by up to 25% and illustrating how forecasting tools are beginning to close the gap between prediction and airport management in practice.

Overall, existing literature shows strong progress in flight delay prediction and its applications. However, there remains a gap in combining predictive modeling with network-aware features and interactive visualization tools that allow users to explore delay patterns across routes and airports in an intuitive and actionable manner.

4. Proposed Method

4.1 Intuition

Flight delays do not occur in a vacuum as an airport's position and connectivity within the aviation network influences its delay characteristics. Traditional models treat flights independently, ignoring this structural context. By modelling the aviation system as a network, we can extract features that reflect an airport's connectivity and centrality, capturing structural dependencies that individual flight-level models overlook.

Furthermore, predicting both delay probability and delay magnitude, a dual prediction target approach, provides a more comprehensive understanding of delay risk as compared to single-target models. Integrating these predictions with interactive visualisations allows users to better interpret and explore delay dynamics across the network.

4.2 Feature Engineering and Data Representation

Our proposed approach introduces route-level feature engineering, where flights are aggregated by origin-destination pairs to capture broader scheduling patterns and reliability trends. To identify meaningful predictors, we investigated the features that had strong correlation with delay outcomes. Pearson correlation was used for continuous variables while the categorical

features like carrier and origin airport were evaluated through group-level aggregation. Categorical variables like origin, destination and carrier were encoded using target encoding, replacing each category with the historical average delay. For example, JFK airport was assigned 18.2 minutes given that flights from JFK were historically delayed by 18.2 minutes on average. Temporal features like flight time were encoded using cyclical encoding to ensure continuity between time boundaries as well.

The aviation network is then represented as a graph where airports are nodes and flights are edges. From this representation, we extract network topology metrics such as node degree, betweenness centrality and edge weights. These features capture structural dependencies within the network and provide additional predictive signals often overlooked by conventional predictive methodologies. We also explored PageRank as an additional network feature to estimate hub influence, but it did not meaningfully improve predictive performance and was excluded from the final model.

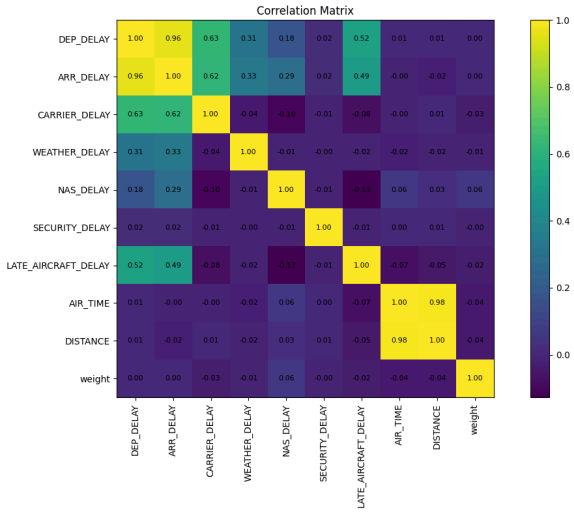

Figure 1: Correlation matrix for delay features

4.3 Predictive Modelling

The prediction task is formulated using two complementary objectives: a classification model to estimate the probability of delay and a regression model to determine expected delay magnitude. This dual-target approach provides a comprehensive risk profile. Given the presence of categorical and time-related data points, tree-based models, such as Decision Trees, Random Forest, XGBoost and CatBoost, were selected and evaluated. A linear regression model was included to establish a baseline of predicting the delay magnitude.

4.4 Visualisation and System Design

Interactive Flight Delay Predictor - This interface allows users to input key flight details and obtain a predicted delay probability and expected delay duration. Each flight is visualised on an interactive map, color-coded by delay risk to allow users to explore delay patterns across routes and airports.

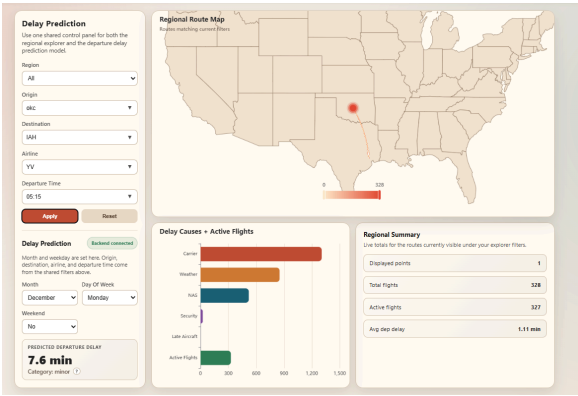

Figure 2: Interactive flight delay predictor

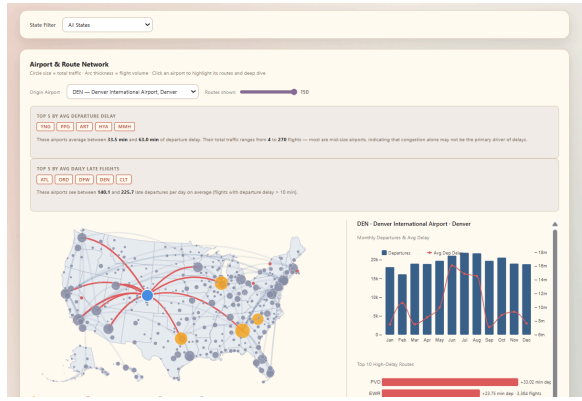

Figure 3: Flight delay insight dashboard

Flight Delay Insight Dashboard - This dashboard provides a comprehensive overview of delay patterns across routes, airlines, airports, and time periods. Users can filter by state and region to explore localised delay trends, identify bottlenecks, and examine how different operational factors relate to departure delay. The dashboard complements the prediction module by enabling exploratory analysis before scenario testing, creating a cohesive end-to-end analytical experience.

Key features include the Airport & Route Network, Delay Heatmap, Carrier Comparison, and Departure Time vs Avg Delay chart, which help substantiate the main features used in modelling. The Airport & Route Network is an interactive visualisation that helps identify airport congestion, spot underperforming routes, and compare airport-level traffic patterns. The Delay Heatmap provides an at-a-glance view of which months and days of the week experience higher delay. Carrier Comparison ranks airlines by average departure delay, highlighting which carriers are more or less reliable. The Departure Time vs Avg Delay Sin chart shows that later departures, especially around the evening period, tend to have higher average delays, suggesting a possible delay snowball effect over the course of the day.

The Regional Explorer (*Figure 2)* extends this analysis with a route map and filter-driven summaries, while the prediction panel allows users to input flight conditions and receive an estimated departure delay category and delay duration. Together, these components support both explanatory analysis and practical forecasting.

5. Evaluation

5.1 Preliminary Data Analysis

The dataset is sourced from Kaggle and contains U.S. domestic flight records for 2018, consisting of over 7 million records. Features include flight schedules, carriers, origin and destination airports, and delays.

Key Observations include:

- Airlines like Frontier (F9), JetBlue (B6), and Allegiant (G6) have higher delays on average, while larger carriers like Southwest (WN) and American (AA) show lower average delays despite handling significantly higher flight volumes.
- High average delays are often associated with low-frequency routes.

- Delays peak on Fridays and during July. Delays are also greater on weekdays rather than weekends.
- Major hubs such as Hartsfield-Jackson Atlanta International Airport (ATL), O'Hare International Airport (ORD), and Dallas/Fort Worth International Airport (DFW) dominate both in traffic volume and network centrality.
- Delay correlations are strongest with late aircraft and airline-related factors.

5.2 Experimental Design

A time-based train-test split was used where earlier months were used for training and later months were used for testing to reflect real-world forecasting conditions and to evaluate the model stability under seasonal variation as well. For delay probability classification, we used Accuracy and ROC/AUC to evaluate predictive performance. For delay duration, we used MAE to measure prediction error.

5.3 Results and Observation

In modelling for the magnitude of the delay, error distributions of all models show a bell-shaped curve with a right skew which suggests slight overestimation. MAE values are 20.77 for Linear Regression, 20.46 for Random Forest, and 20.40 for XGBoost, which performed the best. Additionally, when predicting the delay probability, XGBoost remains the most performant across all 3 ensemble tree-based models explored, in terms of model accuracy and ROC/AUC value.

Figure 4: Error distribution estimation

Ensemble Model	Accuracy	ROC/AUC
XGBoost	0.80	0.74
CatBoost	0.79	0.69
RandomForest	0.79	0.66

Table 1: Model accuracy and ROC/AUC

Given XGBoost's performance, additional SHAP analysis indicates that FL_DATE_MONTH and OP_CARRIER are key drivers of delay magnitude, reflecting seasonal demand and airline operations. For instance, August had the highest average delays, followed by July and then June. While IS_WEEKEND had the smallest impact, we still retained it in the model to ensure we account for busy weekend travel surges. Similarly, when predicting the delay probability, similar features emerge as strong predictors, such as the scheduled flight time, CRS_DEP_TIME, and season factors captured in FL_DATE_DAY and FL_DATE_MONTH.

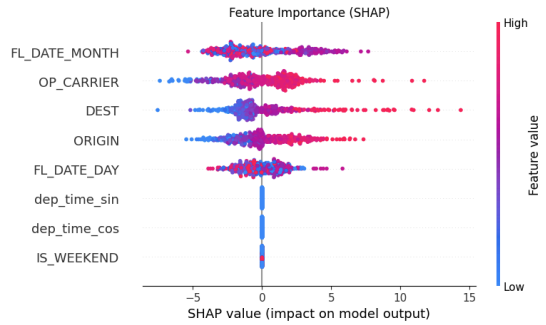

Figure 5: SHAP beeswarm plot for XGBoost (delay magnitude)

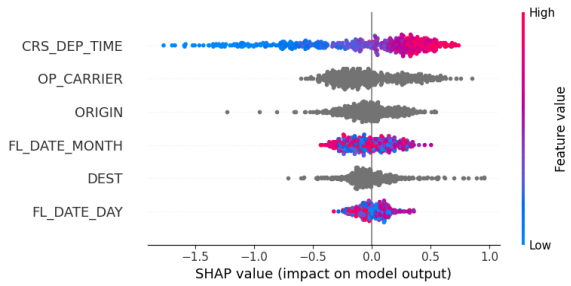

Figure 6: SHAP beeswarm plot for XGBoost (delay probability)

6. Conclusions and Discussions

This project demonstrates the value of incorporating network-level context into flight delay prediction rather than treating flights as isolated events. Network-based features such as node degree and betweenness centrality were incorporated to capture structural context and were identified as predictive signals identified by SHAP analysis. Using a dual-target approach, XGBoost achieved the strongest performance for both delay probability (accuracy: 0.80, ROC/AUC: 0.74) and delay magnitude (MAE: 20.40 minutes). By integrating predictive modelling with interactive visualisation, the system provides users with both accurate predictions and intuitive tools to explore delay patterns across routes and airports, supporting more informed decision-making.

However, the study is limited by the use of a single-year dataset and does not fully capture long-term trends or extreme disruptions. Additionally, while network topology features capture the structural role of airports, the current approach does not directly model temporal delay propagation, an important direction for future work, alongside incorporating real-time data. Overall, the proposed system provides a practical framework for understanding and exploring flight delays, with applications for travelers, airlines and airport authorities.

All team members have contributed a similar amount of effort.

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